

CSE-102: Introduction to Analysis of Algorithms

Lecture Notes

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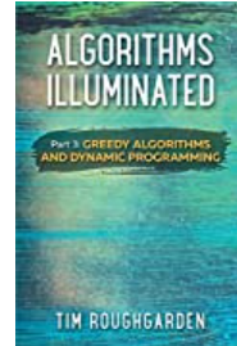
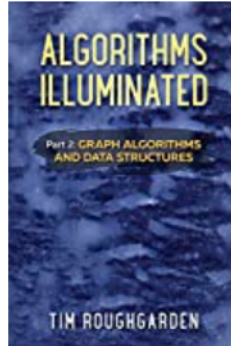
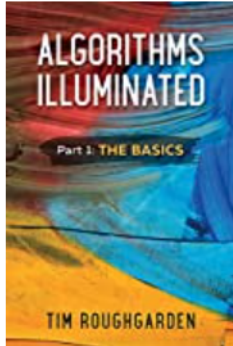
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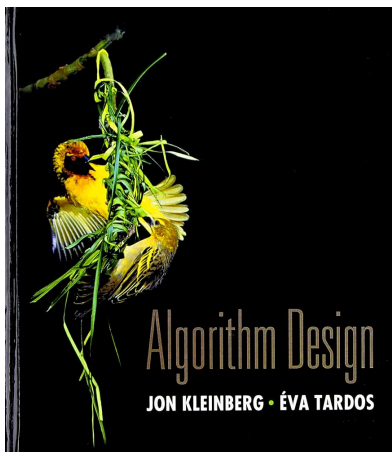
Introduction

Textbooks

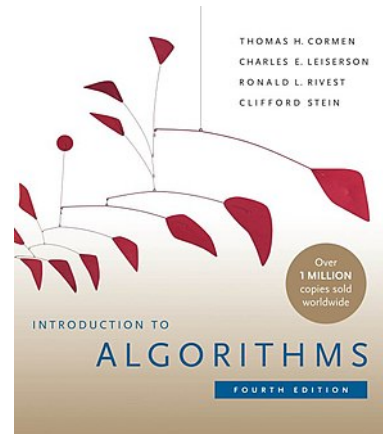
1. Algorithms Illuminated, Vols 1,2 and 3 by Tim Roughgarden.



2. "CLRS" Introduction to Algorithms by Cormen, Leiserson, Rivest and Stein
3. "Algorithm Design" by Kleinberg and Tardos



Algorithm Design by Kleinberg and Tardos



Introduction to Algorithms
by Cormen, Leiserson, Rivest and Stein

What is this Doc?

This doc is my "notes" for the CSE102 course offered at UCSC. I will divide the doc based on each Lecture. For each lecture I shall write down all the important concepts and ideas discussed and their relevant explanations, further I might also add on extra information/elaboration from other sources, thereby making this a comprehensive study resource for me.

Lecture 1:

Etymology of "Algorithms"

- Persian mathematician Muhammad ibn Musa al-Khwarizmi (around 780–850 CE) was a 9th-century scholar, he wrote a book about how to multiply with Arabic numerals.
- Al-Khwarizmi → Algorismi → Algorisme
- When his works were translated into Latin in the 12th century, his name was Latinized to Algoritmi.
- Originally, "Algorisme" [old French] referred to just the Arabic number system, but eventually it came to mean "Algorithm".
- Al-Khwarizmi's work also gave us the word "algebra", which comes from "al-Jabr," a term used in the title of his book.

Integer Multiplication Problem

Problem: Integer Multiplication

Input: Two n -digit nonnegative integers, x and y .

Output: The product $x \cdot y$.

The Integer Multiplication Algorithm will transform the Input to the Output.

- The way we will assess the performance of this algorithm is through the number of basic(or "primitive") operations that are performed.
- "Basic/Primitive/Atomic Operations": Add or Multiply 2 single digit numbers together.
- We will count the number of these primitive operations performed by the following multiplication algorithms as a function of the number of digits, n .

1. Grade-School Multiplication Algorithm

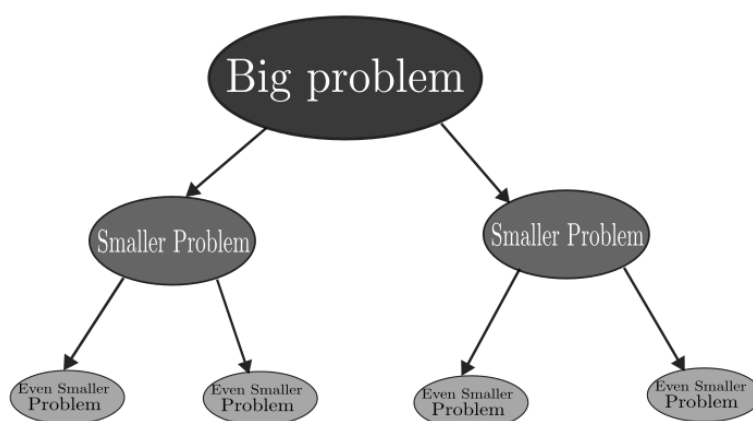
$$\begin{array}{r} \overbrace{123392570752752384623764\dots283568364918374523856298}^n \\ \times 562323582342395285623467\dots235019130750135350013753 \\ \hline \end{array}$$

- Assume you are multiplying 2 n -digit numbers(if one of the numbers doesn't have n -digits you can add leading 0 to make them so).
- Each digit of the bottom number multiplies all the n -digits of the top number, hence a total of n single digit multiplication operations are done, further since these multiplications can cause a carry digit, there can be **at most** n carry digits(even though the first 1-digit multiplication never has carry, don't forget the last digits carry) that need to be added to the product of the single digits
 $\hookrightarrow$ So the total number of Primitive operations performed for each digit of the bottom number is $n + n = 2n$.
- Since the bottom number has n digits, each of those digits will perform $2n$ operations, giving a total of $n \times 2n = 2n^2$ primitive operations performed to obtain all the partial products.

$$\begin{array}{r}
 123392570752752384623764\dots283568364918374523856298 \\
 \times 562323582342395285623467\dots235019130750135350013753 \\
 \hline
 \overbrace{\hspace{10em}}^{2n} \\
 \left. \begin{array}{l}
 \boxed{\hspace{10em}} \\
 \boxed{\hspace{10em}} \\
 \vdots \\
 \boxed{\hspace{10em}} \dots
 \end{array} \right\} n \\
 + \boxed{\hspace{10em}} \dots
 \end{array}$$

- We still need to add the n partial products produced to get the final answer. So we will count the number of single digit atomic additions that are performed, The number of digits in each partial product must equal the number of digits in the last partial product (this is achieved by padding the other partial products with 0's at their start) which is equal to $(n + 1) + (n - 1) = 2n$, each one digit number multiplied by the top number produces $n + 1$ digits and $n - 1$ zeros must be added at the end as part of the algorithm.
 - $\hookrightarrow$ Each partial product has $2n$ digits and there are n of them, hence a total of $2n \times n = 2n^2$ operations will be performed in computing the sum of all the partial products (think of this as adding all the 1-digit numbers in each column, each column has n digits hence n additions must be done and there are $2n$ columns).
- So $2n^2$ operations are performed to compute all the partial products and $2n^2$ operations are done in adding them all. A total of $\boxed{4n^2}$ operations are performed to compute the product of two n -digit numbers using the grade-school algorithm.
- The grade-school algorithm is said to have a time complexity of n^2 [†]. By this we are measuring how the runtime scales with the size of the input. We assume each of the atomic operation take the same time.

2. Divide and Conquer Algorithms for Multiplication



Divide and Conquer

- We now look for a better algorithm for multiplying two n -digit numbers. We shall use the technique of **Divide and Conquer**.
 - $\hookrightarrow$ The idea being, breaking a big complicated problem into smaller(easier) sub-problems. It is often done **recursively**.
Example: Binary Search, Merge Sort.

[†] Later Lecture we see the precise time complexity is $\Theta(n^2)$

2.1 A Recursive Algorithm

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Lecture 2:

3. Karatsuba integer multiplication

Formal Defn of Algorithm Intro of Computational model: RAM model How to describe algorithms
Proving correctness using Loop invariant and/or PMI -¿ Insertion Sort using LI -¿ Merge Sort using
Induction

Lecture 3:

Asymptotic notations. Comparing Asymptotic quantities. Mathematical notations Use of asymptotic notations in equations

Lecture 4:

Recursion tree method

Lecture :

Add the Matrix stuff.

Quick Sort